

# THERMO-DYNAMICS

Reference Book: P.K. Nag,  
C.P. Arora

The aim of study thermodynamics is to convert disorganised form of energy (heat) into organised form of energy (work) in an efficient manner.

### Thermodynamics:

It is the science of energy transfer and its effect on the properties of the system.

### System:

It is a region in space upon which the study is focussed.

### Surroundings:

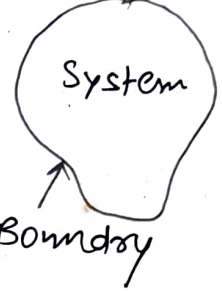
Any external to the system is known as surroundings.

### Boundary

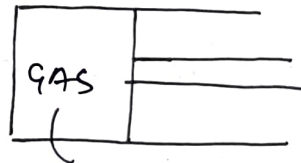
The separation b/w system and surrounding is known as boundary.

Boundary can be either rigid or movable.

Surrounding



|| Rigid  
Boundary



flexible boundary

### Types of system

- closed system
- open system
- isolated system

| System                       | Energy transfer | mass transfer | Example                                                |
|------------------------------|-----------------|---------------|--------------------------------------------------------|
| Closed<br>(Non-flow process) | ✓               | X             | Piston cylinder w/o valves                             |
| Open<br>(Flow process)       | ✓               | ✓             | Turbines<br>pumps<br>compressors                       |
| Isolated                     | X               | X             | universe<br>hot coffee<br>in a well<br>insulated flask |

## Microscopic and macroscopic approach of Thermodynamics

In microscopic approach the behaviour of individual molecule is taken into consideration and this approach is also known as statistical thermodynamics. This approach is useful at low densities i.e. at higher altitude.

In macroscopic approach, the behaviour of individual molecule is not taken into consideration but average behaviour of molecule is taken into consideration. This approach is known as classical thermodynamics.

## Thermodynamic equilibrium

A system is said to be in thermodynamic equilibrium if it satisfies

1. Thermal equilibrium (Equality of temperature)
2. Mechanical equilibrium (Equality of forces/pressure)
3. Chemical equilibrium (Equality of chemical potential)

## Property of a system

Properties are characteristic of a system.

Ex. pressure, volume, temperature etc.

Properties are of two types—

Intensive (Intrinsic)

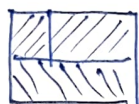
Extensive (Extrinsic)

Intensive properties are independent of mass & size.  
 Ex. Pressure, temperature, density, viscosity,  $K$ , etc.

Extensive properties are dependent on mass or size of the system.

Ex. volume, All forms of energy,

Note: 1. ratio of two extensive property is an intensive property



2. All specific properties are intensive properties.

Ex. specific volume ( $v$ ), specific enthalpy ( $h$ ), specific internal energy ( $u$ ) etc,

$$T \begin{array}{|c|c|} \hline T & T \\ \hline \hline T & \\ \hline \end{array}$$

Intensive

$$V \begin{array}{|c|c|} \hline v_1 & v_2 \\ \hline \hline v_3 & \\ \hline \end{array}$$

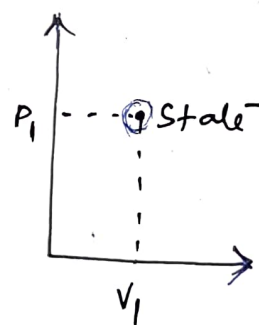
Extensive

Intensive  $\left\{ \begin{array}{l} \text{sp volume } \frac{V}{m} \\ \text{" enthalpy } \frac{H}{m} \\ \text{" energy } \frac{U}{m} \end{array} \right.$

### State of a system

Important - point w.r.t to properties

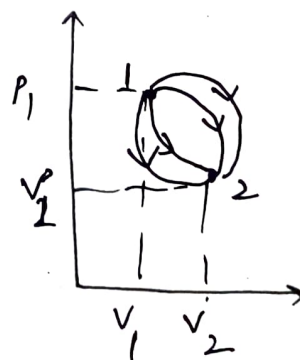
1. Properties are point-functions or state functions.



2. Properties do not depend upon past history

3. Properties are exact differential

\* Condition of a system is known as state of system





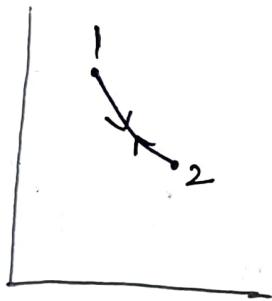
## process

A change of state is known as a process.

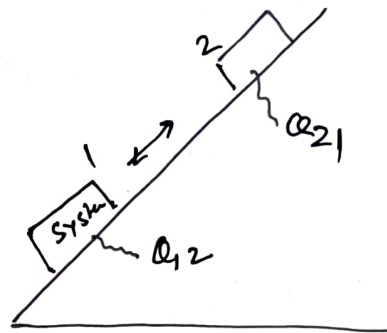
## Reversible and Irreversible process

A process is said to be a reversible process if it reversed in direction followed the same path w/o leaving the any affect on system and surroundy.

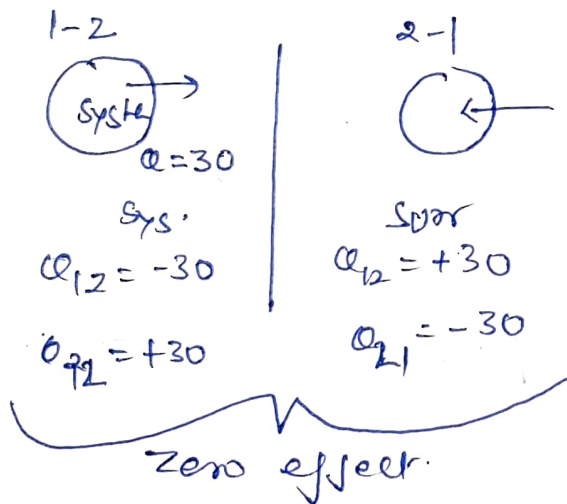
A process which is not a reversible process is an irreversible process. Reversible process is the most efficient process.



Reversible

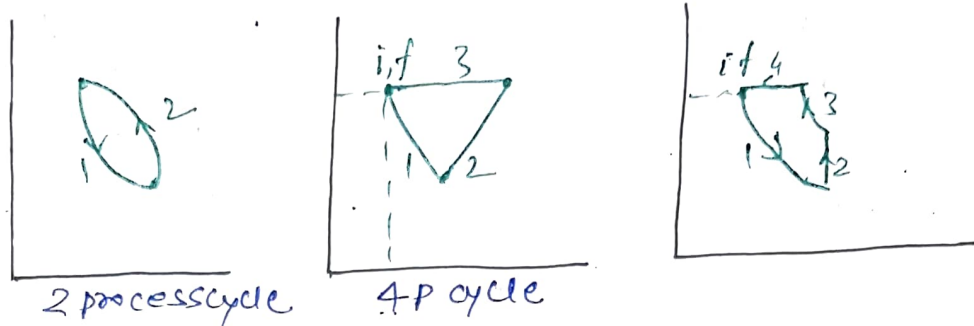


Irreversible



## Thermodynamics cycle

A system is said to <sup>have</sup> undergone a cycle if the initial and final points are same. For a thermodynamic cycle change in property is zero because the i & f points are same.



## Quasi-static process

A process which is carried out in a slow manner with small gradient is known as Quasi-static process. Forthmore Quasi-static process is reversible process.

## Pure substance

A substance is said to be a pure substance if it has a constant chemical composition throughout the volume.

Eg: ice + water, water + water vapour etc



## Gibbs phase rule

According to Gibbs phase rule

$$P + F = C + 2$$

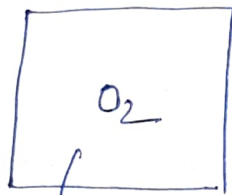
where,  $P$  = no of phases

$F$  = minimum no. of independent intensive variable required to fixed the state (degree of freedom)

$C$  = No of Component.

Eg:

(1)

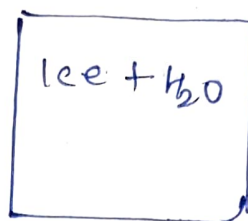


$$P=1, C=1$$

$$P+F=C+2$$

$$F=2 \checkmark$$

(2)



$$P=2, C=1$$

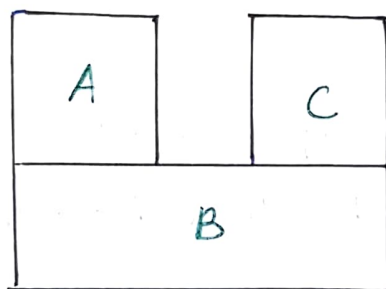
$$P+F=C+2$$

$$\checkmark F=1$$

## ZEROth LAW OF THERMODYNAMICS

This law give the concept of temperature.

Statement: When a body A is in thermal equilibrium with Body B and body B is in thermal equilibrium with body C separately, then <sup>body</sup> A and C are in thermal equilibrium



$$T_A = T_B$$

$$T_B = T_C$$

$$\Rightarrow T_A = T_C$$

In Zeroth law of thermodynamics, one body acts as thermometer

The property which helps in finding temperature is known as thermometric property

### Types of thermometers:

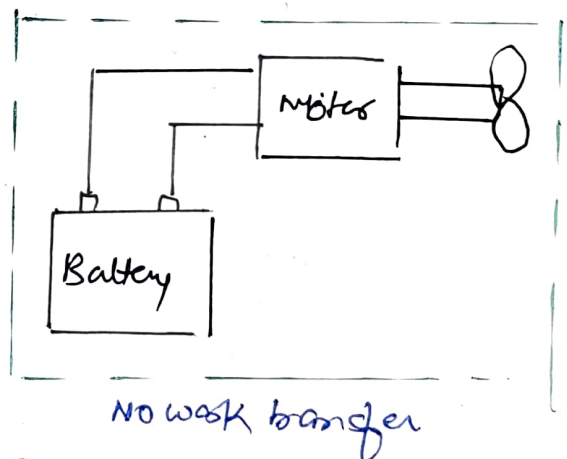
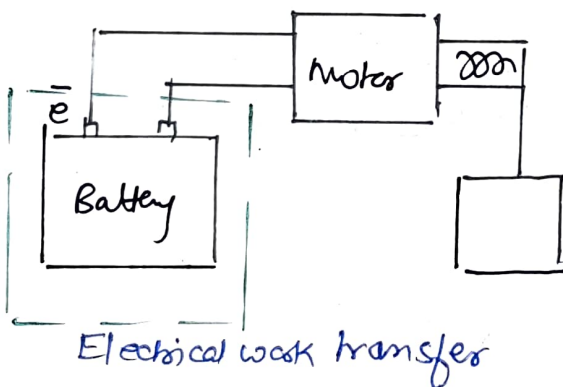
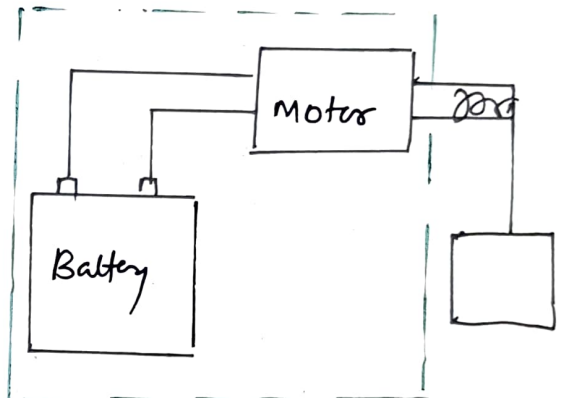
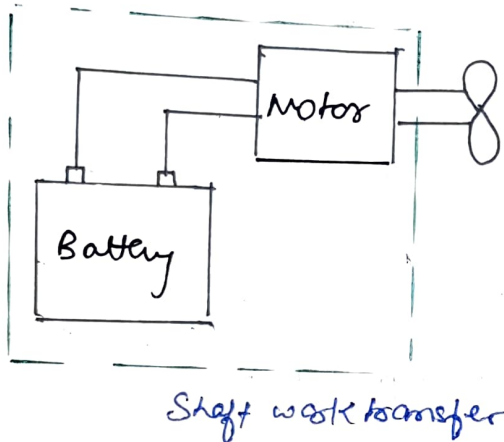
#### 1. Resistance thermometer (Thermistor)

In resistance thermometer, resistance plays the role of thermometric property and it is based on wheatstone bridge principle.

# ENERGY INTERACTIONS (WORK AND HEAT)

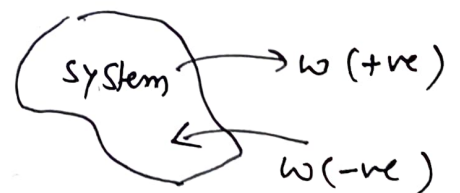
## WORK:

Work is said to be done by the system if the sole effect (complete) on things external to system can be equated to raising of weights (weight may <sup>not</sup> be actually raised)



## CONVECTION OF WORK TRANSFER

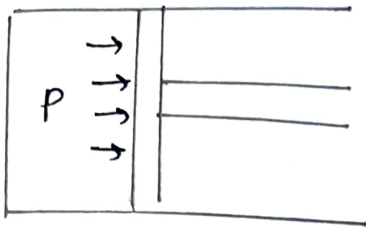
Work done by the system is taken as positive and work done on the system is taken as negative



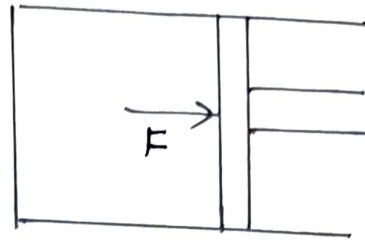


# CLOSED SYSTEM (NON FLOW WORK)

$$P = \frac{F}{A}$$



$$F = PA$$



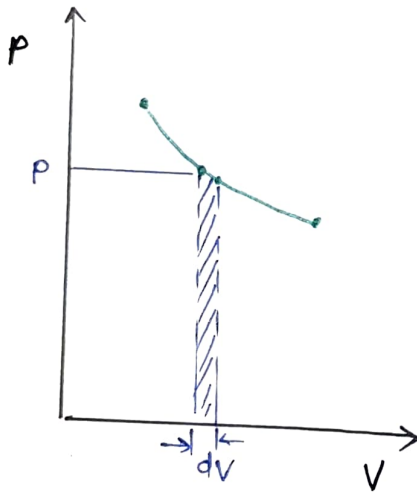
$$P = \frac{F}{A}$$

$$dw = F dx$$

$$dw = PA dx$$

$$dw = P dv$$

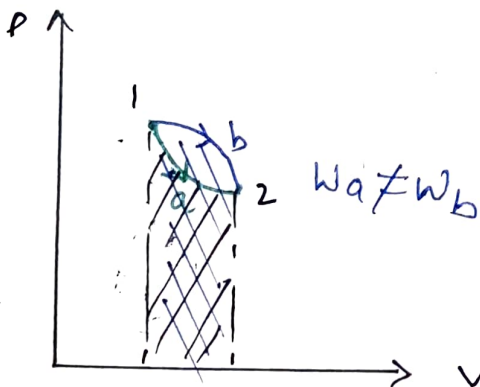
$$W = \int P dv \text{ Closed system work}$$



Area of strip =  $P dv$   
&  $P dv = \text{closed system work}$

$$\underline{\text{Work} = \text{Area of strip}}$$

Area under the curve when projected on volume axis gives closed system work



Though the end points are same for path a & b, areas are different under different curves and hence work transfer is different for different path. Therefore work transfer is a path function.

## Important point w.r.t work transfer

1. Work transfer is a boundary phenomenon.
2. It is a path function (not a property)
3. It is an inexact differential ( $\delta W$  or  $dW$ )

## Condition for applying the equation $W = \int P dV$

1. The system must be a closed system

2. Work should cross the boundary

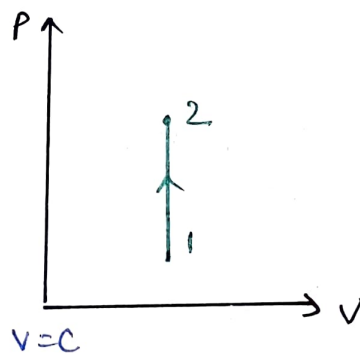
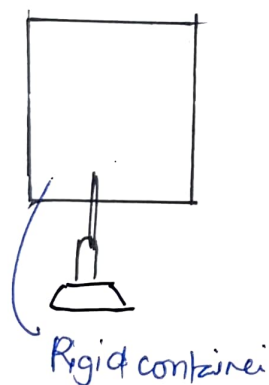
\*\* 3. Process must be reversible process

$P dV$  represent closed system reversible work

## Closed system work (Non flow work) for various processes

1. Constant volume process

(Isochoric or Isomeric process)



$$W = \int P dV$$

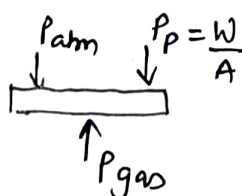
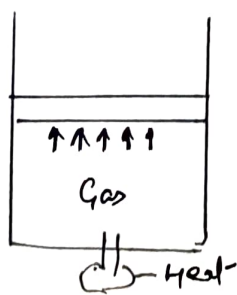
$$dV = 0$$

$$\& W = 0$$

Constant volume closed system work is equal to zero

2. Constant pressure process

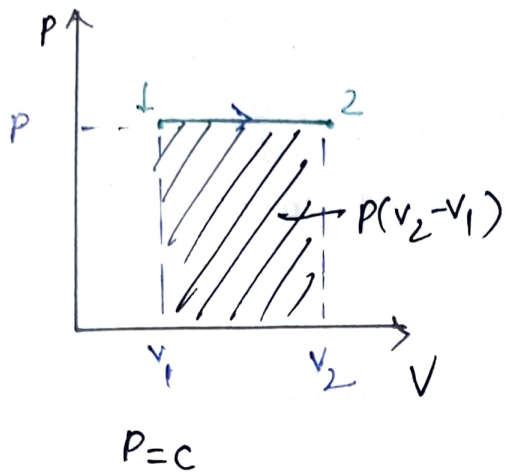
(Isobaric or Isopiestic process)



$$P_{atm} + P_P = P_{gas}$$

$$P_{atm} + \frac{W}{A} = P_{gas} = (\text{constant})$$

As  $P_{atm}$ ,  $W$  &  $A$  are constant,  $P_{gas}$  is also constant



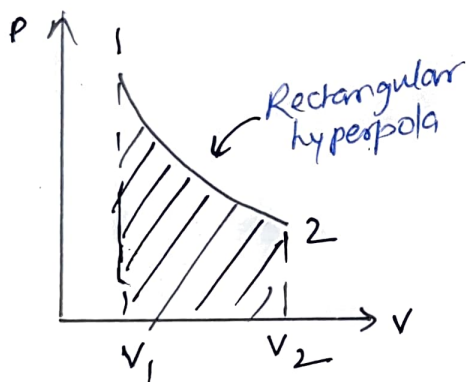
$$W = \int_{V_1}^{V_2} P dV$$

$$W = P(V_2 - V_1)$$

$$W = P(V_f - V_i)$$

### 3. Isothermal process

(Constant temperature process)



Ideal gas

$$PV = nRT$$

$$T = C$$

$$PV = C$$

$$P = \frac{C}{V}$$

$$P_1 V_1 = P_2 V_2 = C = nRT$$

$$W = \int P dV$$

$$W = \int_{V_1}^{V_2} \frac{C}{V} dV$$

$$W = C \ln \frac{V_2}{V_1}$$

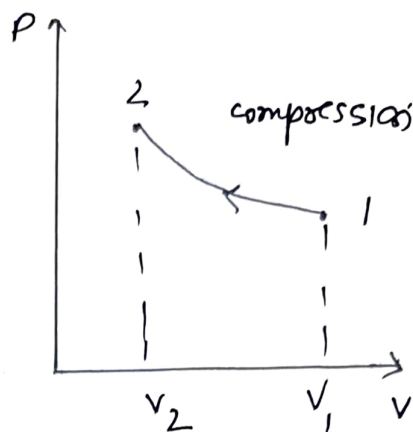
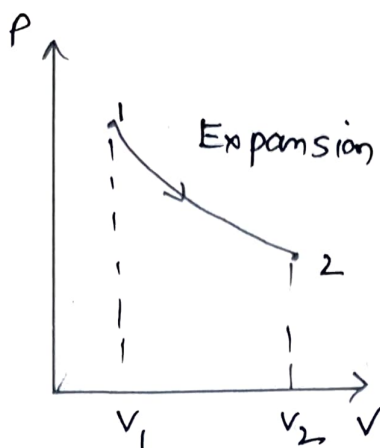
$$\checkmark W = P_1 V_1 \ln \frac{V_2}{V_1}$$

$$\text{As } P_1 V_1 = P_2 V_2$$

$$\checkmark W = nRT \ln \frac{V_2}{V_1}$$

$$W = nRT \ln \frac{P_1}{P_2}$$

$$\checkmark W = P_1 V_1 \ln \frac{P_1}{P_2}$$



$$W = nRT \ln \frac{V_2}{V_1}$$

$$V_2 > V_1$$

$$\frac{V_2}{V_1} > 1$$

$$\ln \frac{V_2}{V_1} = +ve$$

$$\& W = +ve$$

work is done by the system

$$V_2 < V_1$$

$$\frac{V_2}{V_1} < 1$$

$$\ln \frac{V_2}{V_1} = -ve$$

$$W = -ve$$

work is done on the system

#### 4. Adiabatic process

It is the process in which there is no heat transfer from the system <sup>or</sup> to the system.

For an adiabatic process of an ideal gas

$$PV^\gamma = C$$

where  $\gamma$  is adiabatic index and is greater than one  $\gamma > 1$

$$W = \int P dV \rightarrow$$

$$PV^\gamma = C$$

$$P = \frac{C}{V^\gamma} = CV^{-\gamma}$$

$$\therefore P_1 V_1^\gamma = P_2 V_2^\gamma = C$$

$$P_1 = C V_1^{-\gamma}$$

$$P_2 = C V_2^{-\gamma}$$

$$W = \int_{V_1}^{V_2} P dV$$

$$= \int_{V_1}^{V_2} C V^{-\gamma} dV$$

$$= C \left[ \frac{V^{-\gamma+1}}{-\gamma+1} \right] dV$$

$$= \frac{C}{-\gamma+1} \left[ V_2^{-\gamma+1} - V_1^{-\gamma+1} \right]$$

$$= \frac{[P_2 V_2 - P_1 V_1]}{-\gamma+1}$$

$$W = \frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$$



## 5. Polytropic process

A process in which  $PV^\eta = C$  is valid is known as polytropic process. For this process generally  $\eta > 1$ ;  $\eta < \gamma$  ( $1 < \eta < \gamma$ ).

In polytropic process there is both heat transfer and work transfer but whereas in adiabatic process there is only work transfer and there is no heat transfer.

$$\text{Polytropic work} = \frac{P_1 V_1 - P_2 V_2}{\eta - 1}$$

General form:  $PV^K = C$

$$P = C$$

$$PV^K = C$$

$$K = 0$$

$$P = C$$

$$V = C$$

$$PV^K = C$$

$$(P V^K)^{1/K} = C^{1/K}$$

$$P^{1/K} V = C$$

$$K = \infty$$

$$V = C$$

Process

K

|    |            |              |
|----|------------|--------------|
| 1. | $P = C$    | 0            |
| 2. | $V = C$    | $\infty$     |
| 3. | $T = C$    | $K = 1$      |
| 4. | Adiabatic  | $K = \gamma$ |
| 5. | Polytropic | $K = \eta$   |

$$T = C$$

$$PV = nRT$$

$$PV = C$$

$$PV^K = C$$

$$K = 1$$

Adiabatic

$$PV^\gamma = C$$

$$PV^K = C$$

$$K = \gamma$$

Polytropic

$$PV^\eta = C$$

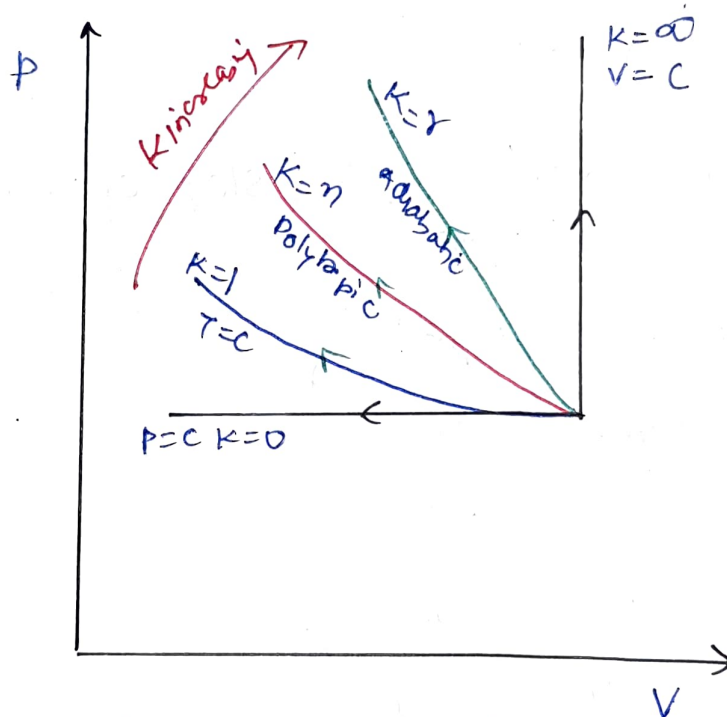
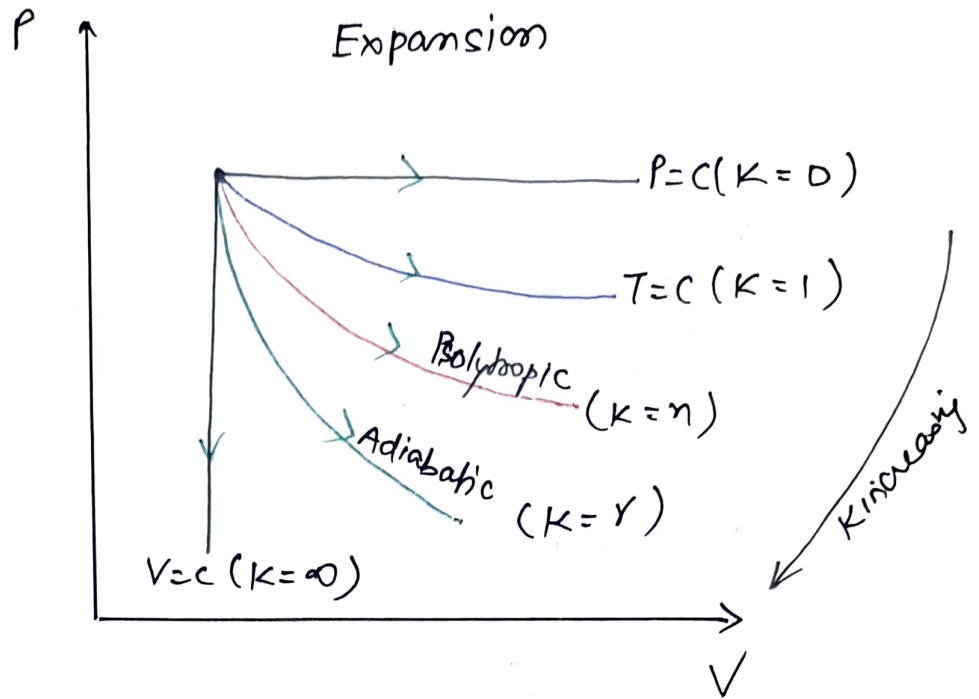
$$PV^K = C$$

$$K = \eta$$

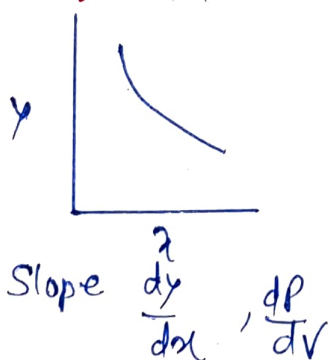
ascending order of K

|            |              |
|------------|--------------|
| $P = C$    | $K = 0$      |
| $T = C$    | $K = 1$      |
| Polytropic | $K = \eta$   |
| Adiabatic  | $K = \gamma$ |
| $V = C$    | $\infty$     |





Slope of isothermal curves on P-V diagram



For isothermal process ( $T=C$ )

$$PV = mRT$$

$$PV = \text{constant}(C)$$

$$Pdv + vdp = 0$$

$$Pdv = -vdp$$

$$\frac{dp}{dv} = -\frac{P}{V}$$

(Slope of isothermal curve)

# Ideal gas equations for various processes:

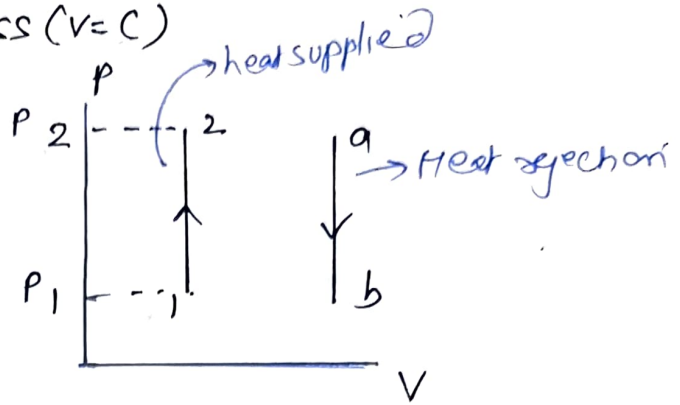
## 1. Constant volume process ( $V=C$ )

$$V=C$$

$$PV = nRT$$

$$P \propto T$$

$$** \frac{P_2}{P_1} = \frac{T_2}{T_1}$$



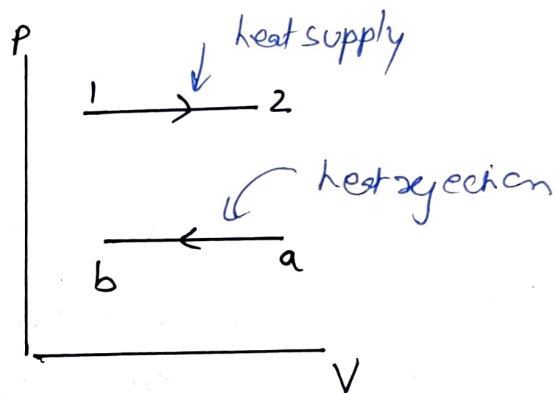
## 2. Constant pressure process ( $P=C$ )

$$P=C$$

$$PV = nRT$$

$$V \propto T$$

$$** \frac{V_2}{V_1} = \frac{T_2}{T_1}$$



## 3. Isothermal process ( $T=C$ )

$$T=C$$

$$PV = nRT$$

$$PV = \text{Constant}$$

$$** P_1 V_1 = P_2 V_2$$

## 4. Adiabatic process

$$PV^\gamma = C$$

$$** P_1 V_1^\gamma = P_2 V_2^\gamma$$

$$PV = nRT$$

$$P = \frac{nRT}{V}$$

$$\frac{nRT}{V} V^\gamma = C$$

$$T \cdot V^{\gamma-1} = C$$

$$** T_1 V_1^{\gamma-1} = T_2 V_2^{\gamma-1}$$

$$** \frac{T_2}{T_1} = \left( \frac{P_2}{P_1} \right)^{\frac{\gamma-1}{\gamma}}$$

Valid for ideal gas

5. polytropic process

$$P_1 V_1^\eta = P_2 V_2^\eta$$

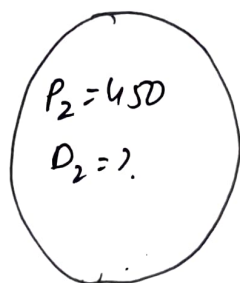
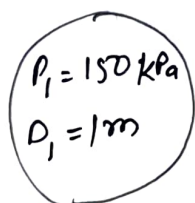
$$T_1 V_1^{\eta-1} = T_2 V_2^{\eta-1}$$

$$\frac{T_2}{T_1} = \left( \frac{P_2}{P_1} \right)^{\frac{\eta-1}{\eta}}$$

valid for polytropic  
process of an  
ideal gas

Prob 1 A spherical balloon of 1 m diameter contains a gas at 150 kPa. The gas inside the balloon is heated until the pressure reaches 450 kPa. During the process of heating. The pressure of gas inside the balloon is proportional to cube of the diameter of balloon. Find (1) Work done by the gas inside the balloon

Sol<sup>n</sup> The given system is a closed system and let us assume the process to be reversible process.



$$P \propto D^3$$

$$P = K D^3$$

$$150 = K (1)^3$$

$$K = 150$$

$$W = \int P dV$$

$$W = \int K D^3 dV$$

$$W = \int 150 D^3 dV$$

$$= \int 150 D^3 \frac{\pi D^2 dD}{2}$$

$$= \frac{150 \pi}{2} \int_{D_1}^{D_2} D^5 dD$$

$$V = \frac{4}{3} \pi R^3$$

$$= \frac{4}{3} \pi \left( \frac{D}{2} \right)^3$$

$$V = \frac{\pi D^3}{6}$$

$$dV = \frac{\pi D^2 dD}{2}$$



$$= \frac{150 \pi}{12} [D_2^6 - D_1^6]$$

$$= \frac{150 \pi}{12} [3^2 - 1^6]$$

$$= \frac{150 \pi}{12} \times 8$$

$$= 100 \pi$$

$$W = 314.15 \text{ KJ}$$

Since  $W$  is  $+$ ve, so work is done by the system